Hunter Huffman

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EDUCATION

The University of Toledo, Toledo, OH

Bachelor of Science: Mechanical Engineering, *GPA 3.43* — Achieved: December 2024
COBI Minor: Entrepreneurship, Family, and Small Business, *GPA 4.00* — Achieved: Spring 2023
Master of Science: Mechanical Engineering, GPA 3.76 — Anticipated Graduation Date: May 2026
Master of Business Administration: MBA Finance, *GPA 3.86* — Anticipated Graduation Date: May 2026

HONORS & AWARDS

The University of Toledo, Toledo, OH

Dean's List: Spring 2021, Summer 2022, and Spring 2024
President's List: Fall 2024

EXTRACURRICULARS

University of Toledo Rocketry Club – Member — January 2026 – Present

- Owner of the implementation of solar cells on the payload to power electronics. CAD design and technical drawings for the payload
- Hands on assembly of the payload and other components. Provided technical advice for undergrad students within the club

CERTIFICATIONS

Six Sigma DMAIC Green Belt – Credential ID HH15567GB — Achieved: July 2025

- Implementing DMAIC (Define, Measure, Analyze, Improve, and Control) methodologies in various areas within a company that can enhance processes, reduce defects, and improve product quality

L1 Rocketry Certification — Achieved: March 2026

- Ability to design, assemble, and fly a rocket with a class H or I rocket motor and safely land it with no major damage.

TECHNICAL SKILLS

- AutoCAD 2025
- Plant3D 2023
- Navisworks 2023
- Fusion 360
- SolidWorks 2025
- MATLAB
- Microsoft Office Suite
- Adams View
- Inventor 2025
- 3D Printing
- Writing, Analyzing, and Performing Test Plans
- Design Failure Mode and Effect Analysis (DFMEA) Process
- Hands On Technical Work
- Jump and Jump Scripting (JMP)
- Bluebeam Revu
- Technical Drawings
- Data Analysis
- Financial Modeling
- LabView
- Octave
- Siemens Programmable Logic Controller (PLC)
- Lathe, Milling, Grinding, and Drilling
- Operating Power Tools

Relevant Coursework

- Thermodynamics
- Fluid Mechanics
- Heat Transfer
- Mechanical Design
- Additive Manufacturing
- Automation Controls
- Mechatronics
- Materials Science
- Reliability
- Six Sigma
- Vibrations
- Design of Experiments
- Manufacturing Processes
- MATLAB

RELEVANT ENGINEERING CAREER EXPERIENCE

First Solar, Perrysburg, OH

Development Engineer December 2024 – Present

- Managed one direct report and provided technical oversight as well as decision making.
- R&D on glass and glass quality focusing mostly on the production process and large defects over 1-2 mm^3.
- Material testing and analysis with plastics, polymers, and glass. Performed tensile, impact, 3 and 4 point bending, UV degradation, corrosion, and thermal cycling tests.
- Executed and created test plans for DFMEA derisk. Decreased 300-400 risk priority number (RPN) areas to 100-200 RPN.
- Assisted reliability and various development groups on testing, custom manufacturing builds, and test analysis.
- Troubleshoot problems and hands-on technical work with the development tools and other machines.
- Assisted in the design and creation of a new development tool with a pneumatic cannon. Tool had a custom chronograph to measure velocity and could reach above 80 J in impact energy.
- Performed statistical analysis and report outs on completed tests. Including, Weibull distribution curves from completed hail tests to obtain P90 energies for shot and module pass rates.
- Designed components to optimize current development testing tools. For instance, designed and built a lift system for the technicians to be able to check for glass breakage underneath a module during testing.
- JMP scripting and toolkit to extract and analyze data.
- Created a design of experiment to evaluate glass thickness (2.0 mm to 2.8 mm) vs glass strength (4000 psi to 8000 psi).
- Assisted in creating business cases for current and future products in development to understand cost impact.

Senior Design Project, The University of Toledo

Subsonic Wind Tunnel - Full Scale and Half Scale Senior Semester/Fall 2024

Project description: The goal of the project was to fully design a full-scale subsonic wind tunnel and produce a working half-scale subsonic wind tunnel. The reason behind the project was to promote the aerospace major and minor that was being added to the degree program list at the University of Toledo. The wind tunnels will be used in the aerospace and fluids labs for various labs and research.

Experience:

- Conducted Solidworks modeling for both wind tunnels.
- Performed Solidworks CFD and stress analysis. Simulation showed a velocity of roughly 50 m/s for the air flow in the test section, validating the prediction and calculations for the full-scale model.
- Fluids design and calculations. Converted from turbulent to laminar flow using mesh screens, honeycombs, and turning vanes.
- Performed calculations to obtain specifications on different parts of the wind tunnel, including the ambient and static pressure of the wind tunnel, fan CFM, and size of the fan and turning vanes.
- Aided in the design process and drafting of the technical drawings for the 25.6 ft x 9.9 ft x 5.6 ft full-scale wind tunnel.
- 3D printed and assembled the 6.4 ft x 2.7 ft x 1.4 ft half-scale wind tunnel.

Internship Experience

First Solar - Hail Development Engineering Intern-hired part-time after rotation May 2024 – December 2024

- Took over the DFMEA for a newly designed product that is as an add on to solar panels in respect to hail.
- Created and executed test plans based on DFMEA development parameters.
- Utilized JMP and Excel to analyze data sets and compile into various types of plots.
- Fixing and writing JMP script to extract data.
- Troubleshoot problems and hands-on technical work with development tools.

Rudolph Libbe Group - VDC/Mechanical Intern-hired part-time in between full-time rotation January 2022 – May 2024

- Created over 100 CAD orthographic drawings for the field and fabrication shop.
- Designed various parts on CAD based on 2D sketches and drawings.
- Assisted project managers to bid large >$1,000,000 projects and do take offs.
- Used Leica P20, Matterport, and Trimble Unit to perform site scans.
- Used Real Works and Cyclone to tie scans together for modeling using point clouds.